

## **Study of Food Waste Composition and Generation from Campus Food Vendors: A Case Study of Chiang Mai University**

Chadaporn Sriwichai<sup>1</sup>, Santi Phithakkitnukoon<sup>2</sup>, Karn Patanukhom<sup>3</sup> and Sarunnoud Phuphisith<sup>4\*</sup>

<sup>1</sup>Graduate student, Department of Environmental Engineering, Faculty of Engineering, Chiang Mai University, Chiang Mai 50200, Thailand; <sup>2</sup>Associate Professor, Department of Computer Engineering, Faculty of Engineering, Chiang Mai University, Chiang Mai 50200, Thailand; <sup>3</sup>Assistant Professor, Department of Computer Engineering Faculty of Engineering, Chiang Mai University, Chiang Mai 50200, Thailand; <sup>4</sup>Assistant Professor, Department of Environmental Engineering, Faculty of Engineering, Chiang Mai University, Chiang Mai 50200, Thailand

\*Phone : 053 944 192 , E-mail : sarunnoud.p@cmu.ac.th

### **ABSTRACT**

This study aims to evaluate food waste generation and its composition from the cooking process of curry and rice menus to promote sustainable production and consumption. Data were collected from ten food vendors located at cafeterias in Chiang Mai University for two working day weeks each, using the direct weighing method. The results showed that food waste was generated the most during the preparation stage. Daily food waste generation ranged from 0.53 – 5.17 kg and 0.002-0.029 kg/plate sold. Avoidable and unavoidable ratios differed by shop. Components of food waste varied depending on the menu and the practices of each vendor. None of the surveyed shops separated food waste, making it difficult to use to its maximum benefit.

**Keywords :** Food Waste; Avoidable; Unavoidable; Waste management; Thai Cuisine

### **INTRODUCTION**

Food is one of the most essential factors for human consumption to sustain life. When food is produced, it often generates food waste; More than one third of food produced is waste [1], affecting food scarcity and environmental issues. Food waste may be occurred from leftover food or food scraps that occur in various steps, from harvesting to cooking. All food waste has an impact on the economy and the environment, including hunger, food shortages, resource waste and greenhouse gas emissions from food waste management. Therefore, studying ways to reduce and prevent food waste is an important issue in promoting sustainable consumption, which will be beneficial in reducing production costs, reducing negative environmental impacts and reducing unnecessary resource waste.

Food waste can be generated from many sources, such as households, wholesale and retail stores, hotels, restaurants, and others. Each source may have different amounts and components of food waste [2] which are affected by the size, the type of cooking, the efficiency of stock management and kitchen plan, and the behavior of operators. Food waste can be divided into avoidable and unavoidable. Avoidable refers to food waste that is still edible, which may be leftovers, discarded, or unused. It may come from trimming during the preparation of ingredients before cooking or during the cooking process, such as carrot scraps that are trimmed to be used as a garnish on a dish, or meat scraps, etc. [3]. Unavoidable refers to food-related components that are not intended for human consumption. Examples of inedible parts include bones, shells, and seeds. Unavoidable parts do not include packaging [4].

The food service sector is one of the major sources of food waste, with many factors contributing to food waste generation. The study by Filimonau et al. (2023) [5] reported that the size and type of restaurant affect the amount of food waste generated, and the composition of food waste generated varies from type to type. The efficiency of restaurant management, kitchen management plans, and the behavior of restaurant operators, such as planning for ingredient preparation, checking product labels before purchasing to avoid expiration before use and having to discard, purchasing ingredients in excess, etc.

The present study focused on the generation of in-kitchen waste in Thai cuisine. We investigated the amount and composition of food waste generated from the cooking process, covering stocking ingredients, preparing, cooking and ready to serve. We also surveyed the current food waste management practices to promote food waste reduction and preventive measures.

## METHODOLOGY

This study targeted curry and rice food vendors as this type of food category uses various ingredients and processes. Food waste generation and composition data were collected from ten food vendors located at cafeterias in Chiang Mai University for two working day weeks each, beginning on October 4, 2024, and ending on January 31, 2025. The direct weighing method was employed and food waste was classified into avoidable and unavoidable categories. Food waste was also accessed by ingredient types, i.e., vegetable scraps, meat, starch, and eggshells. In addition, the operators were asked about their cooking experiences, steps in producing food waste and how it was managed. All data was analyzed using Microsoft Excel version 2019.

## RESULTS AND DISCUSSIONS

### 1. Food waste composition and generation rate

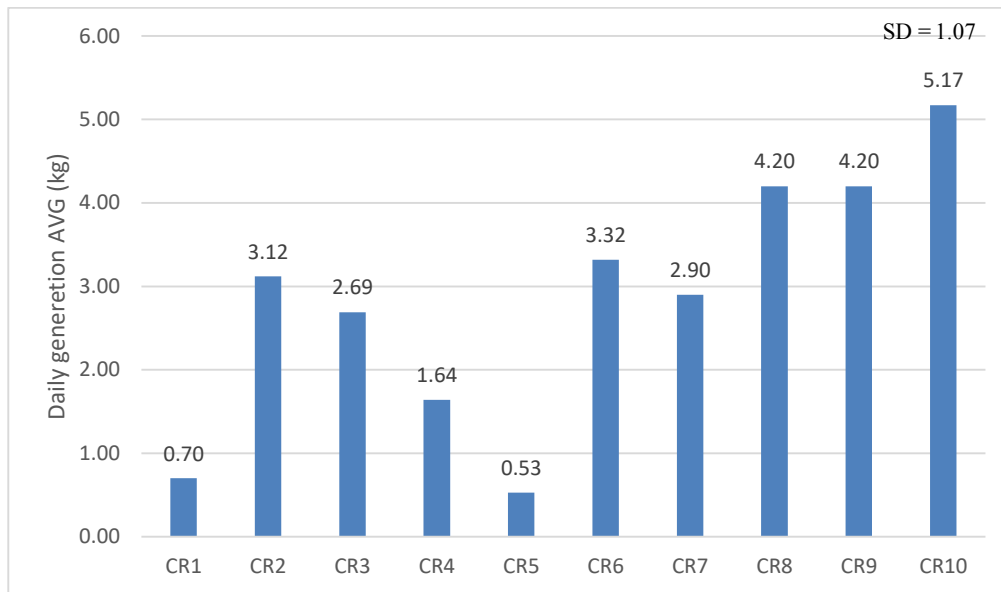
General data of the food vendors surveyed are shown in Table 1. About 75% of the food vendors had more than 10 years of cooking experience. Most of them prepared and cooked all meals in the cafeteria kitchen

**Table 1** General data of the surveyed food vendors

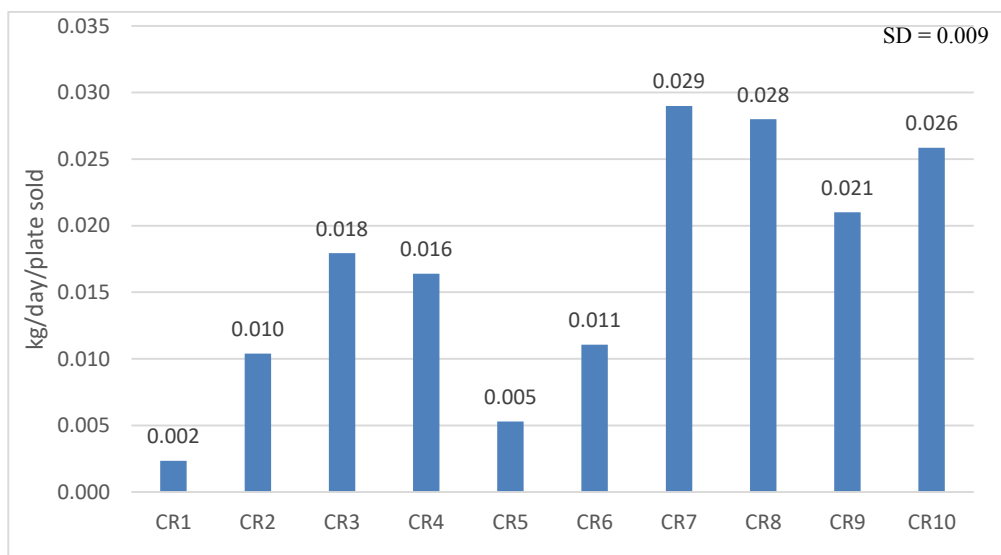
| ID   | Cooking Experience (years) | Plate sold (per/day) | Cooking at cafeteria kitchen* (%) |
|------|----------------------------|----------------------|-----------------------------------|
| CR1  | 20                         | 300                  | 100                               |
| CR2  | 20                         | 300                  | 100                               |
| CR3  | 20                         | 150                  | 20                                |
| CR4  | 10                         | 100                  | 50                                |
| CR5  | 2                          | 100                  | 100                               |
| CR6  | 20                         | 300                  | 100                               |
| CR7  | 13                         | 100                  | 20                                |
| CR8  | 15                         | 150                  | 50                                |
| CR9  | 20                         | 200                  | 20                                |
| CR10 | 30                         | 200                  | 20                                |

\*Some food vendors prepared and cooked at cafeteria only, some food vendors partly prepared and cooked at their houses.

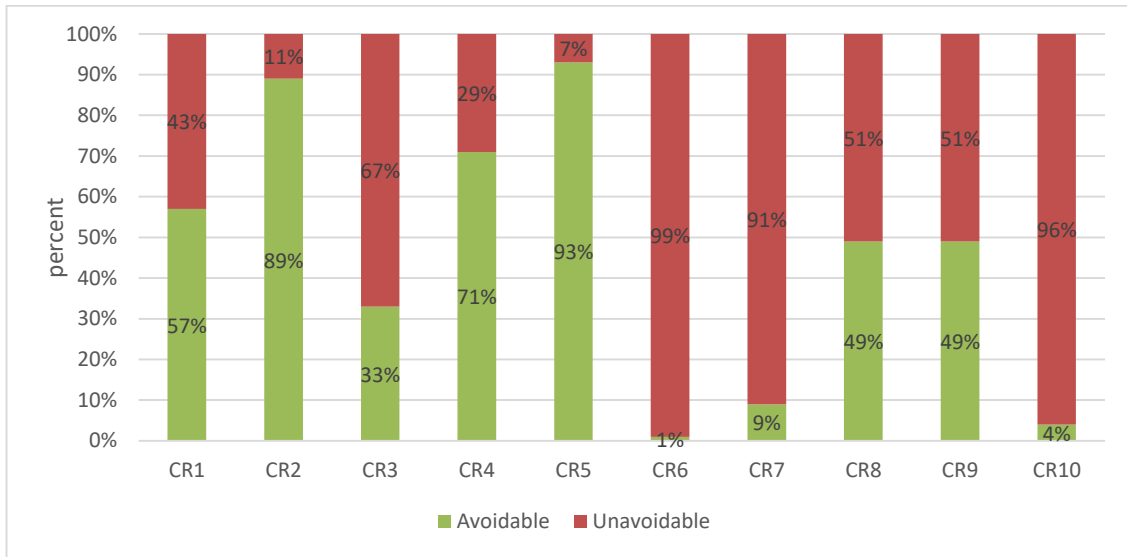
The average amount of daily food waste generation ranged from 0.53 – 5.17 kg, as shown in **Figure 1**. Considering the amount of plate sold daily, the food waste generation varied from 0.002-0.029 kg/day/plate sold, as shown in **Figure 2**. It is noticeable that sales volume does not affect the amount of food waste generated per day by curry and rice food vendors. The shops CR1, CR2, and CR6 had the same amount of daily plate sold, but the amount of daily food waste generated was different. The shop CR1 had less food waste generation because the food vendors used their trimmed vegetable scraps to cook other menus instead of throwing them away. The CR2 was a popular shop for selling boxed lunches, resulting in a higher amount of food waste generated per day than both shops. Shops CR3 and CR8 had the same plate sold, but CR8 had higher amount of food waste generation because all cooking processes were done at the university cafeteria kitchen. The shop CR3, on the other hand partly prepared and cooked from outside. Shops CR4, CR5, and CR7 had the same plate sold, but the amount of food waste was different. CR5 has the least amount of food waste because the food vendors buy and sell trimmed ingredients, so the amount of food waste is less than the other two food vendors.



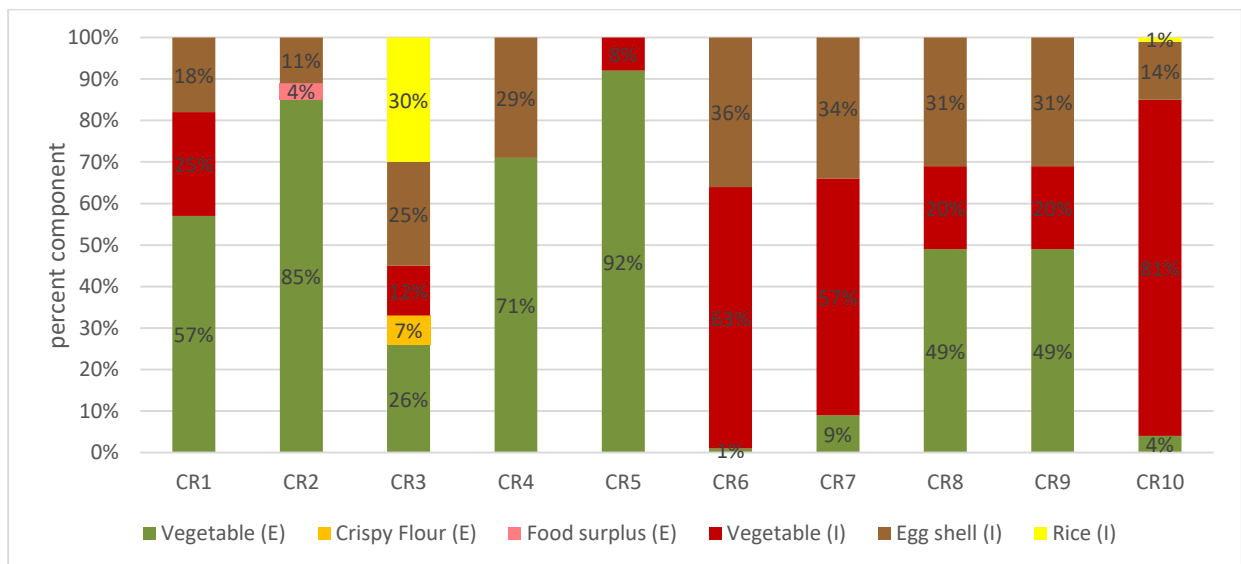
**Figure 1** The daily average amount of food waste generation



**Figure 2** The daily average amount of food waste generation per plate sold



**Figure 3** Avoidable and unavoidable compositions in food waste



**Figure 4** Component details in food waste

In average, the preparation of ingredients was the highest source of food waste generation, accounting for 95% of the total waste generated. The second source of food waste occurred in cooking process (4%), and ready-to-serve (1%). This is consistent with previous research [5]. The composition between avoidable and unavoidable parts varied, as shown in **Figure 3**. About half of the surveyed shops generated unavoidable parts over avoidable parts. The dominant was vegetable scraps, which could be avoided, at 44% on average, unavoidable vegetable scraps, at 36%, eggshells, at 25%, burnt rice scraps, at 16 %, crispy dough scraps at 8%, and ready-to-serve food, at 4%, as shown in **Figure 4**.

## 2. Food waste management methods

All the surveyed shops did not separate food waste and disposed of it mixed with general waste. The shops mentioned about the limited kitchen space; placing several bins for separation was impossible and separating waste takes time. Most vendors (93%) threw away all the generated food waste at the university bins while only 7% separated their food waste for composting.

## CONCLUSION

The present study assessed food waste generation from curry and rice cooking. The preparation stage was the key source of food waste generation. Food vendors generated unavoidable more than avoidable parts, with vegetables being the most commonly found component. Components of food waste varied depending on the menu and the practices of each vendor. None of the surveyed shops separated food waste, resulting in food waste mixed with general waste, making it difficult to use to its maximum benefit.

## ACKNOWLEDGEMENT

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